

$$1) a) \sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n}$$

Ratio test: $\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \lim_{n \rightarrow \infty} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)(2n+1)}{2 \cdot 4 \cdot \dots \cdot 2n(2n+2)} \cdot \frac{2 \cdot 4 \cdot \dots \cdot 2n}{1 \cdot 3 \cdot \dots \cdot (2n-1)}$

$$= \lim_{n \rightarrow \infty} \frac{2n+1}{2n+2} = 1 \text{ inconclusive} \Rightarrow$$

$\Rightarrow$ R-D test.

$$\lim_{n \rightarrow \infty} n \left(\frac{x_n}{x_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{2n+2}{2n+1} - 1 \right) =$$

$$= \lim_{n \rightarrow \infty} n \cdot \left(1 + \frac{1}{2n+1} - 1 \right) = \frac{1}{2} < 1 \Rightarrow$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n} \text{ is DIVERGENT.}$$

$$c) \sum_{n=1}^{\infty} a^{h_n}, \quad a > 0$$

Ratio test: $\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \lim_{n \rightarrow \infty} \frac{a^{h(n+1)}}{a^{h_n}} = \lim_{n \rightarrow \infty} a^{h(n+1) - h_n} =$

$$= \lim_{n \rightarrow \infty} a^{h \frac{n+1}{n}} = a^{h \cdot 1} = a^h = a^0 = 1 \text{ "inconclusive"} \Rightarrow$$

$$\Rightarrow \text{R-D test: } \lim_{n \rightarrow \infty} n \left(\frac{x_n}{x_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} n \left(a^{\frac{n}{n+1}} - 1 \right)$$

$$= \lim_{n \rightarrow \infty} n \cdot \frac{a^{\frac{n}{n+1}} - 1}{\frac{n}{n+1}} \cdot \frac{n}{n+1} = -\ln a \quad \text{Recall: } \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$$

$$\begin{aligned} \lim_{n \rightarrow \infty} n \cdot \ln \frac{n}{n+1} &= \lim_{n \rightarrow \infty} \ln \left(\frac{n}{n+1} \right)^n = \ln \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \\ &= \ln \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right)^n = \ln \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right)^{-(n+1) \cdot n \cdot \frac{-1}{n+1}} = \\ &= \ln e^{-1} = -1 \end{aligned}$$

Since the limit depends on a , we discuss over a :

$$-\ln a < 1 \Leftrightarrow \ln a > -1 \Leftrightarrow a > \frac{1}{e} \Rightarrow \text{CONV.}$$

$$0 < a < \frac{1}{e} \Rightarrow \text{DIV}$$

$$a = \frac{1}{e} \Rightarrow \sum_{n=1}^{\infty} e^{n/n} = \sum_{n=1}^{\infty} n \text{ which is DIV.}$$

Raabe - Duhamel Test

$$\lim_{n \rightarrow \infty} n \left(\frac{x_n}{x_{n+1}} - 1 \right) = L \begin{cases} > 1 : \text{CONV} \\ < 1 : \text{DIV} \end{cases}$$

1) a), c) at the table (will redo above)

$$1) \quad d) \quad \sum \frac{a^n \cdot n!}{n^n} \quad , \quad a > 0$$

$$\text{ratio test: } \lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \lim_{n \rightarrow \infty} \frac{a^{n+1} \cdot (n+1)!}{(n+1)^{n+1} n^n} \cdot \frac{n^n}{a^n \cdot n!} =$$

$$= \lim_{n \rightarrow \infty} \frac{a \cdot n^n}{(n+1)^n} = \lim_{n \rightarrow \infty} a \cdot \left(\frac{n}{n+1} \right)^n =$$

$$\Rightarrow \frac{1}{e} \text{ (previous ex.)}$$

$$= a \cdot \frac{1}{e} = \frac{a}{e}$$

- if $a < e \Rightarrow \frac{a}{e} < 1 \Rightarrow \text{CONV.}$
- if $a > e \Rightarrow \frac{a}{e} > 1 \Rightarrow \text{DIV.}$
- if $a = e \Rightarrow \frac{a}{e} = 1$ inconclusive $\Rightarrow$

R-D test $\sum \underbrace{\frac{e^n \cdot n!}{n^n}}_{x_n}$

$\lim_{n \rightarrow \infty} n \left(\frac{1}{e} \cdot \left(\frac{n+1}{n} \right)^n - 1 \right)$ complicated:

Recall: $\sum x_n \text{ conv.} \Rightarrow x_n \rightarrow 0$
 $x_n \not\rightarrow 0 \Rightarrow \text{div.}$

check monotony of x_n

$$\downarrow \frac{x_{n+1}}{x_n} = \frac{\cancel{e^{n+1}} \cdot \cancel{(n+1)!}}{(\cancel{n+1})^{n+1}} \cdot \frac{n^n}{\cancel{e^n} \cdot \cancel{n!}} =$$

$$= \frac{e \cdot n^n}{(n+1)^n} = e \left(\frac{n}{n+1} \right)^n > 1$$

$$\left(\frac{n+1}{n} \right)^n = \left(1 + \frac{1}{n} \right)^n \rightarrow e \quad \left(1 + \frac{1}{n} \right)^n \geq e$$

$$\left(\frac{n+1}{n} \right)^n < e$$

$$\Rightarrow x_{n+1} > x_n \Rightarrow \lim_{n \rightarrow \infty} x_n > 0 \Rightarrow \text{div.}$$

$$2) a) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n(n+1)}}$$

Leibnitz test
 $x_n \searrow 0 \Rightarrow \sum (-1)^n \cdot x_n$
 conv.

$$x_n = \frac{1}{\sqrt{n(n+1)}} \rightarrow 0$$

$$\frac{x_{n+1}}{x_n} = \sqrt{\frac{n(n+1)}{(n+1)(n+2)}} = \sqrt{\frac{n}{n+2}} < 1$$

x_n decreasing $\Rightarrow$
 $x_n \rightarrow 0$

$$2b. \Rightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}} \text{ conv.}$$

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{\sqrt{n(n+1)}} \right| = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}}$$

$$\frac{\frac{1}{\sqrt{n(n+1)}}}{\frac{1}{n}} = \frac{n}{\sqrt{n(n+1)}} \rightarrow 1 \Rightarrow$$

$\Rightarrow$ the two series which were compared above have the same nature $\Rightarrow$

$$\Leftrightarrow \frac{1}{\sqrt{n(n+1)}} \text{ "like" } \frac{1}{n} \int \Rightarrow$$

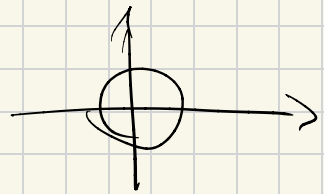
$$\sum \frac{1}{n} \text{ Div}$$

$$\Rightarrow \sum \frac{1}{\sqrt{n(n+1)}} \text{ Div.}$$

$$b) \sum (-1)^n \cdot \sin \frac{1}{n}$$

$$x_n = \sin \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \sin \frac{1}{n} = \sin 0 = 0$$



$$\frac{1}{n} \in [1, 0)$$

$$\sin \text{ increasing on } [0, \frac{\pi}{2}]$$

$$\frac{1}{n+1} < \frac{1}{n}$$

$$\sin \frac{1}{n+1} < \sin \frac{1}{n} \Rightarrow x_n \searrow$$

⇒

conv. by the Leibnitz test.

$$\sum_{n \geq 1} \sin \frac{1}{n}$$

$$\frac{\sin x}{x} \xrightarrow{x \rightarrow 0} 1$$

$$\frac{\sin \frac{1}{n}}{\frac{1}{n}} \rightarrow 1 \Rightarrow \sum_{n \geq 1} \sin \frac{1}{n} \text{ "like" } \sum_{n \geq 1} \frac{1}{n}$$

can be any constant (one is equal to the other multiplied by a const.)

Comparison test

$$\sum \frac{1}{n} \text{ Div.} \Rightarrow \sum \sin \frac{1}{n} \text{ Div.}$$

c) $\sum_{n \geq 1} \frac{\sin n}{n^2}$

$$0 < |\sin n| \leq 1$$

fun fact of the day

$$\frac{\pi^2}{6}$$

$$\sum_{n \geq 1} \left| \frac{\sin n}{n^2} \right| = \sum_{n \geq 1} \frac{|\sin n|}{n^2} < \sum_{n \geq 1} \frac{1}{n^2} \Rightarrow$$

CONV.

$$\stackrel{C.T}{\Rightarrow} \sum_{n=1}^{\infty} \frac{|\ln n|}{n^2} \text{ CONV. } \Rightarrow$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{\ln n}{n^2} \text{ absolutely convergent } \Rightarrow \text{ CONV.}$$

$$3) \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots = \frac{1}{1-x}$$

Radius of convergence $R = 1$

The series converges absolutely for $|x| < 1$

$\Rightarrow$ we can differentiate it term by term

$$\sum_{n=1}^{\infty} (x^n)' = 0 + 1 + 2x + \dots = \sum_{n=1}^{\infty} n x^{n-1} =$$

$$= \left(\frac{1}{1-x} \right)'$$

$$\left(\frac{1}{u} \right)' = -\frac{u'}{u^2}$$

$$\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}$$

$$\sum_{n=1}^{\infty} n \cdot x^n = \frac{x}{(1-x)^2}$$

• Differentiate $\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}$

$$\sum_{n=2}^{\infty} n(n-1) \cdot x^{n-2} = \frac{2}{(1-x)^3} \quad | \cdot x^2$$

$$\sum_{n=2}^{\infty} n(n-1) x^n = \frac{2x^2}{(1-x)^3}$$

4) $\sum_{n=0}^{\infty} t^n = 1 + t + t^2 + \dots = \frac{1}{1-t} \quad |t| < 1$

$$\int_0^x dt : \sum_{n=0}^{\infty} \int_0^x t^n dt = \int_0^x \frac{1}{1-t} dt$$

$$\sum_{n=0}^{\infty} \frac{t^{n+1}}{n+1} \Big|_0^x = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = -\ln(1-t) \Big|_0^x =$$

$$\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x)$$

$$\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x), \quad |x| < 1$$

$$\sum_{n=0}^{\infty} (-t)^n = \frac{1}{1+t}, \quad |t| < 1$$

$$\sum_{n=0}^{\infty} (-1)^n \cdot t^n = \frac{1}{1+t}, \quad |t| < 1 \quad \left| \int_0^x dt \right|$$

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} = \ln(1+x)$$

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = \ln(1+x)$$

OR: $x \rightarrow -x \Rightarrow \sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^n}{n} = \ln(1+x) \cdot (-1)$

$$5) \sum_{n=1}^{\infty} (-1)^n t^n = \frac{1}{1+t}, \quad |t| < 1$$

$$t \rightarrow t^2$$

$$\sum_{n=0}^{\infty} (-1)^n t^{2n} = \frac{1}{1+t^2} \quad \left| \int_0^x dt \right.$$

$$\sum (-1)^n \frac{x^{2n+1}}{2n+1} = \arctan t \Big|_0^x = \arctan x, \quad \text{for } |x| < 1 \quad (\text{as } |t| < 1).$$

$$x=1: \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n+1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$$

by the Leibnitz test

$\arctan 1 \rightarrow x=1.$